

Q1.

Ainee is the owner of several car shops and employs multiple managers who oversee the daily operations. She stores the data about her business in a relational database named "CARS." The database has three tables: SHOP, MANAGER, and CAR.

The SHOP table contains columns for ShopID, ManagerID, Address, Town, and TelephoneNumber.

The MANAGER table has columns for ManagerID, FirstName, LastName, DateOfBirth, and Wage.

The CAR table has columns for RegistrationNumber, Make, Model, NumberOfMiles, and ShopID.

a) Identify which fields in each table are primary keys and which ones are foreign keys.

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[2]

b) Explain how access rights can be used to safeguard the data in Ainee's database against unauthorized access.

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[3]

c) (i) Ainee uses DDL and DML statements in her database. Complete the following DML statements:

To retrieve the number of cars for sale in each shop.

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SELECT ....., COUNT (.....) AS NumberOfCars
FROM .....
..... ShopID;
```

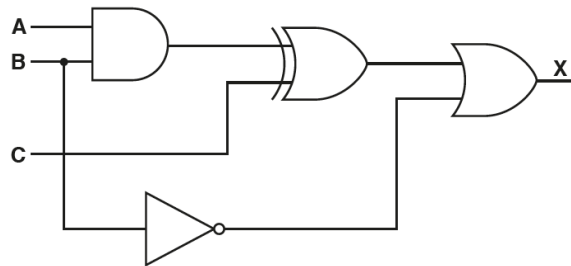
[3]

(ii) To insert a new record for a car with the following data: RegistrationNumber: 123AA, Make: Tiger, Model: Lioness, NumberOfMiles: 10500, ShopID: 12BSTREET.

```
..... (RegistrationNumber, Make, Model, NumberOfMiles, ShopID)
..... ('123AA', 'Tiger', 'Lioness', 10500, '12BSTREET');
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[2]**Q2.**

A logic circuit is shown:



(a) Write the logic expression for the logic circuit.

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[3]

(b) Complete the truth table for the given logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

c) Identify one logic gate not used in the given logic circuit. Draw the symbol for the logic gate and complete its truth table.

Logic gate:

Symbol:

Truth table:

A	B	Output
0	0	
0	1	
1	0	
1	1	

[3]

Q3.

The following table shows part of the instruction set for a processor that has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Op code	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC.
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC.
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC.
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the Index Register. Copy the contents of this calculated address to ACC.
LDR	#n	Immediate addressing. Load the number n to IX.
STO	<address>	Store the contents of ACC at the given address.
STX	<address>	Indexed addressing. Form the address from <address> + the contents of the Index Register. Copy the contents of ACC to this calculated address.
ADD	<address>	Add the contents of the given address to ACC.
INC	<register>	Add 1 to the contents of the register (ACC or IX).
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX).
JMP	<address>	Jump to the given address.
CMP	<address>	Compare the contents of ACC with the contents of <address>.
CMP	#n	Compare the contents of ACC with number n.
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True.
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False.
IN		Key in a character and store its ASCII value in ACC.
OUT		Output to the screen the character whose ASCII value is stored in ACC.
END		Return control to the operating system.

Study the following low level language code and answer subsequent questions related to the same. These instructions are using three arrays (Array1, Array2 & Array3), which are saved in memory locations 100, 200 and 300 respectively.

	LDR #0	; load initial value to IX
	LDM #300	; load the address of first element of Array3 into C
	STO C	
	LDM #0	
	STO D	; initiate loop counter
LOOP:	LDX 100	; load content of next element of Array1 into A
	STO A	
	LDX 200	; load content of next element of Array2 into B
	STO B	
	LDD A	; load content of A into accumulator
	ADD B	; add content of B to accumulator
	STX C	; store the result from the indexed memory address stored in C
	LDD D	; loop count
	INC ACC	
	CMP #3	; check if all elements of arrays 1 & 2 are read and added to 3rd
	JPE ENDD	; end if at end of array
	STO D	
	INC IX	; increment pointer to next elements in Array1 & Array2
	JMP LOOP	; repeat loop if not at the end of array
ENDD:	END	; end of program
100	3	
101	4	
102	5	
200	1	
201	2	
202	8	

(a) Dry-run this assembly language program using the trace table.

[illegible]

(b)

(i) What is the purpose of the given assembly language code?

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[4]

(ii) What is the size of the three arrays?

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[1]

(iii) What is the significance of the label "LOOP:" in the code?

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[1]

(iv) What does the instruction "CMP #3" do?

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[1]

(v) Which instruction would you modify, and to what instruction, to subtract the content of location B from A and store the result in address saved in C?

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[1]

Q4:

(a) Give denary equivalents of the following IEC binary prefixes used to represent quantities of digital information:

(i) 1 Kibi:

 [1]

(ii) 1 Mebi:

 [1]

(iii) 1 Gibi:

 [1]

(b) A computer's hard disk has a capacity of $\frac{1}{2}$ **terabytes (TB)**. Convert this capacity to the nearest number of **gibibytes (GiB)**, using the appropriate IEC binary prefix.

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 [2]

(c) Perform the following binary addition using given bits only. Show your working.

(i) 11010000 + 10110000

 [2]

(ii) Determine whether an overflow occurred during the addition. Explain your reasoning briefly.

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[1]

(iii) Convert the result to denary.

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[1]

(iv) Convert the result to hexadecimal.

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[1]

Q5:

ZAK wants to protect the sensitive data on his company's computer network from unauthorized access and malware attacks.

(a) Explain the importance of data encryption and access controls in ensuring data security on a computer network.

Explanation:

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[3]

(b) ZAK is considering implementing antivirus software on his company's computers. Describe how antivirus software works to protect against malware attacks and provide an example of a common type of malware.

Explanation:

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Example:

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[3]

(c) Two common types of cyber-attacks are phishing and pharming. Describe the difference between phishing and pharming.

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[2]

Q6.

Harim is starting her first job as a software developer for a multinational company.

(a) Harim has been advised to join a professional ethical body.

Explain the benefits to Harim of joining a professional ethical body.

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[3]

(b) Harim is shown the software she will be working on. She is unfamiliar with the Integrated Development Environment (IDE) she is required to use.

(i) Describe the ethical ways in which Harim can act in this situation.

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[2]

(ii) In addition to debugging tools, identify three other tools or features commonly found in an IDE that support the writing of a program.

- 1
- 2
- 3

[3]

(c) Harim is part of a team writing a program. She finds an error in part of the program that has already been tested. She decides not to tell anyone because she is worried about the consequences.

Explain the reasons why Harim acted unethically in this situation.

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[2]

(d) Harim's team use language translators.

Complete the descriptions of language translators by writing the missing words.

..... are usually used when a high-level language program is complete. They translate all the code at the same time and then run the program. They produce files that can be run without the source code. translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line. Assemblers are used to translate assembly code into

[4]

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Q7.**(a)** Table below shows four stages in converting sound into a digital form.

Show the correct order for the stages by labelling them with the numbers 1 – 4

(1 being the first stage).

Stage	Order (1-4)
Binary representation of level stored	
microphone picks up sound waves	
value read at specific point and rounded to a level	
converted to an electrical analogue signal	

[3]**(b)** Describe how a black and white image could be represented as a bitmap in binary.

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[3]**(c)** A screen contains black text on a solid white background. One line of the screen might be represented as follows with W representing white pixel and B representing a black pixel.

WWWWWWBBBWWWWWWWWWWWWBBBWWBBBWWWW

A compression technique is applied to the line of data above and results in the following output:

6W3B12W6B1W3B4W

State what data compression algorithm has been applied.

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[1]

(d) Another data compression technique is JPEG, which is a lossy compression algorithm.

Explain what is meant by lossy compression.

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[1]

Q8.

Fill in the blanks.

The two main types of networks are _____ and _____. The key difference between them is that first one is confined to a small geographic area, while the other can cover a larger geographic area, such as a city, country, or even the entire world.

The two main models of networked computers are _____ and _____. In the first model, one computer (the server) provides services or resources to other computers (the clients) on the network. In the other, all computers on the network are equal and can share resources with each other.

The different types of subnetwork models include bus, star, _____, and _____ topologies. Each has its own benefits and drawbacks, such as scalability, reliability, and ease of implementation.

The hardware used in LAN networks includes switches, servers, NICs, WNICs, WAPs, cables, bridges, and repeaters. These devices are used to connect computers and other devices on the network, and facilitate communication between them.

The role of a _____ in a network is to direct data packets between different networks. Ethernet is a set of rules and protocols that govern how data is transmitted over a LAN network, and includes mechanisms to avoid collisions, such as

_____.

_____ is a model for delivering on-demand computing resources over the internet. _____ are available to anyone over the internet, while _____ are reserved for a single organization or business. The benefits include scalability, flexibility, and cost-effectiveness, while drawbacks include security and privacy concerns.

[4]